Parallel Mater is a physics API offering functions and structures that allow you to define physics simulations. Parallel Mater supports Rigid Bodies, Fluid, Cloth, Rope, Smoke, and Soft Bodies. Parallel Mater uses a configured fixed-function pipeline to simulate the specified physics simulations and their specified couplings.

Parallel Mater

Function names: pmCreateInstance
Structure and type names: PmInstanceCreateInfo
Constants and return codes: PM_SUCCESS
PFN_pmName: pointer type for pmName
[n.n] Specification section

Return codes [2]

Success & error codes [2.1]

Return codes are reported via PmResult return values.

```
typedef enum PmResult {
    PM_SUCCESS = 0,
    PM_NOT_READY = 1,
    PM_TIMEOUT = 2,
    PM_INCOMPLETE = 3,
    PM_ERROR_INVALID_ARGUMENT = -1,
    PM_ERROR_INVALID_STATE = -2,
    PM_ERROR_OUT_OF_HOST_MEMORY = -3,
    PM_ERROR_OUT_OF_DEVICE_MEMORY = -4,
    PM_ERROR_FEATURE_NOT_PRESENT = -5,
    PM_ERROR_FORMAT_NOT_SUPPORTED = -6,
    PM_ERROR_CAPACITY_EXCEEDED = -7,
    PM_ERROR_DEVICE_LOST = -8,
    PM_ERROR_INCOMPATIBLE_VERSION = -9
} PmResult;
```

Function Pointers and Instancing [3]

Function pointers [3.1]

```
typedef void (PMAPI_CALL
  *PFN_pmVoidFunction)(
    void);
PFN_pmVoidFunction pmGetInstanceProcAddr(
    PmInstance instance, const char* pName);
PFN_pmVoidFunction pmGetDeviceProcAddr(
    PmDevice device, const char* pName);
```

Instance creation [3.2]

```
typedef struct PmInstanceCreateInfo {
    PmStructureType sType;
    const void* pNext;
    uint32_t apiVersion;
    const char* pApplicationName;
    PmBool32 enableValidation;
} PmInstanceCreateInfo;
PmResult pmEnumerateInstanceVersion(
    uint32_t* pApiVersion);
PmResult pmCreateInstance(
    const PmInstanceCreateInfo* pInfo,
    const PmAllocationCallbacks* pAllocator,
    PmInstance* pInstance);
void pmDestroyInstance(PmInstance instance);
```

Host allocation callbacks [3.3]

```
typedef void* (PMAPI_CALL *PmAllocateFn)(
    void* user, size_t size, size_t
      alignment);
typedef void (PMAPI_CALL *PmFreeFn)(
    void* user, void* allocation);
typedef struct PmAllocationCallbacks {
    void* pUserData;
    PmAllocateFn pfnAllocate;
    PmFreeFn pfnFree;
} PmAllocationCallbacks;
```

Devices & queues [4]

Physical devices [4.1]

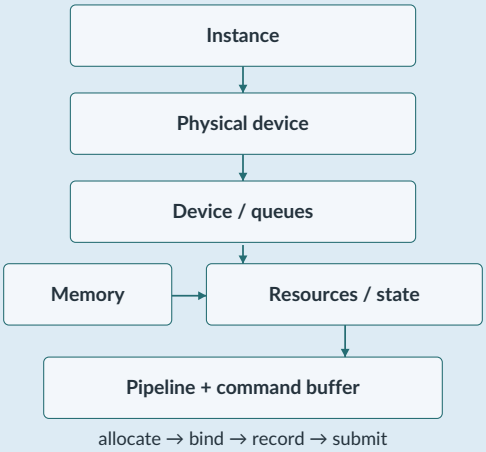
```
PmResult pmEnumeratePhysicalDevices(
    PmInstance instance, uint32_t* pCount,
    PmPhysicalDevice* pDevices);
```

Physical-device identity [4.2]

```
typedef struct PmPhysicalDeviceProperties {
    uint32_t apiVersion;
    uint32_t driverVersion;
    uint32_t vendorId;
    uint32_t deviceId;
    uint8_t deviceUuid[16];
    char deviceName[256];
} PmPhysicalDeviceProperties;
void pmGetPhysicalDeviceProperties(
    PmPhysicalDevice physicalDevice,
    PmPhysicalDeviceProperties*
      pProperties);
```

Queue families [4.3]

```
enum {
    PM_QUEUE_PHYSICS_BIT = 0x1,
    PM_QUEUE_TRANSFER_BIT = 0x2
};
typedef struct PmQueueFamilyProperties {
    PmFlags queueFlags;
    uint32_t queueCount;
    uint32_t timestampValidBits;
    float timestampPeriodNanoseconds;
} PmQueueFamilyProperties;
PmResult pmGetPhysicalDeviceQueueFamilies(
    PmPhysicalDevice physicalDevice,
    uint32_t* pCount,
    PmQueueFamilyProperties* pProperties);
```



allocate → bind → record → submit

Supported vs enabled features [4.4]

```
typedef struct PmPhysicalDeviceFeatures {
    PmBool32 rigidBodies;
    PmBool32 rigidConstraints;
    PmBool32 particleLiquids;
    PmBool32 liquidRigidCoupling;
    PmBool32 liquidLiquidCoupling;
    PmBool32 timelineSemaphores;
    PmBool32 timestampQueries;
    PmBool32 statisticsQueries;
    PmBool32 memoryBudget;
    PmBool32 vulkanInterop;
} PmPhysicalDeviceFeatures;
void pmGetPhysicalDeviceFeatures(
    PmPhysicalDevice physicalDevice,
    PmPhysicalDeviceFeatures* pFeatures);
```

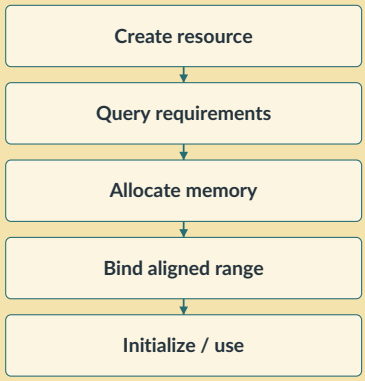
Logical device [4.5]

```
typedef struct PmDeviceQueueCreateInfo {
    uint32_t queueFamilyIndex;
    uint32_t queueCount;
    const float* pPriorities;
} PmDeviceQueueCreateInfo;
typedef struct PmDeviceCreateInfo {
    PmStructureType sType;
    const void* pNext;
    uint32_t queueCreateInfoCount;
    const PmDeviceQueueCreateInfo*
      pQueueInfos;
    PmPhysicalDeviceFeatures
      enabledFeatures;
} PmDeviceCreateInfo;
PmResult pmCreateDevice(
    PmPhysicalDevice physicalDevice,
    const PmDeviceCreateInfo* pInfo,
    PmDevice* pDevice);
void pmDestroyDevice(PmDevice device);
void pmGetDeviceQueue(PmDevice device,
    uint32_t family, uint32_t index,
    PmQueue* pQueue);
```

Explicit device memory [5]

Heaps & types [5.1]

```
enum {
    PM_MEMORY_DEVICE_LOCAL_BIT = 0x1,
    PM_MEMORY_HOST_VISIBLE_BIT = 0x2,
    PM_MEMORY_HOST_COHERENT_BIT = 0x4,
    PM_MEMORY_HOST_CACHED_BIT = 0x8
};
typedef struct PmMemoryHeap {
    PmDeviceSize size;
    PmFlags flags;
} PmMemoryHeap;
typedef struct PmMemoryType {
    PmFlags propertyFlags;
    uint32_t heapIndex;
} PmMemoryType;
typedef struct PmMemoryProperties {
    uint32_t heapCount;
    PmMemoryHeap heaps[16];
    uint32_t typeCount;
    PmMemoryType types[32];
} PmMemoryProperties;
void pmGetPhysicalDeviceMemoryProperties(
    PmPhysicalDevice physicalDevice,
    PmMemoryProperties* pProperties);
```



Allocation ranges

geometry	state	scratch

Requirements & suballocation [5.2]

```
typedef struct PmMemoryRequirements {
    PmDeviceSize size;
    PmDeviceSize alignment;
    uint32_t memoryTypeBits;
    PmBool32 requiresDedicatedAllocation;
} PmMemoryRequirements;
```

Allocate & free [5.3]

```
typedef struct PmMemoryAllocateInfo {
    PmStructureType sType;
    const void* pNext;
    PmDeviceSize allocationSize;
    uint32_t memoryTypeIndex;
} PmMemoryAllocateInfo;
PmResult pmAllocateMemory(PmDevice device,
    const PmMemoryAllocateInfo* pInfo,
    PmDeviceMemory* pMemory);
void pmFreeMemory(PmDevice device,
    PmDeviceMemory memory);
```

Map & host visibility [5.4]

```
typedef struct PmMappedMemoryRange {
    PmDeviceMemory memory;
    PmDeviceSize offset;
    PmDeviceSize size;
} PmMappedMemoryRange;
PmResult pmMapMemory(PmDevice device,
    PmDeviceMemory memory,
    PmDeviceSize offset, PmDeviceSize size,
    void** ppData);
void pmUnmapMemory(PmDevice device,
    PmDeviceMemory memory);
PmResult pmFlushMappedMemoryRanges(
    PmDevice device, uint32_t count,
    const PmMappedMemoryRange* pRanges);
PmResult pmInvalidateMappedMemoryRanges(
    PmDevice device, uint32_t count,
    const PmMappedMemoryRange* pRanges);
```

Buffers & triangle geometry [6]

Buffer creation [6.1]

```
enum {
    PM_BUFFER_TRANSFER_SRC_BIT = 0x1,
    PM_BUFFER_TRANSFER_DST_BIT = 0x2,
    PM_BUFFER_GEOMETRY_INPUT_BIT = 0x4,
    PM_BUFFER_PARTICLE_INPUT_BIT = 0x8,
    PM_BUFFER_SIMULATION_OUTPUT_BIT = 0x10
};
typedef struct PmBufferCreateInfo {
    PmStructureType sType;
    const void* pNext;
    PmDeviceSize size;
    PmFlags usage;
    uint32_t initialQueueFamilyIndex;
} PmBufferCreateInfo;
PmResult pmCreateBuffer(PmDevice device,
    const PmBufferCreateInfo* pInfo,
    PmBuffer* pBuffer);
void pmDestroyBuffer(PmDevice device,
    PmBuffer buffer);
```

Buffer requirements & binding [6.2]

```
typedef struct PmBufferRange {
    PmBuffer buffer;
    PmDeviceSize offset;
    PmDeviceSize size;
} PmBufferRange;
void pmGetBufferMemoryRequirements(
    PmDevice device, PmBuffer buffer,
    PmMemoryRequirements* pRequirements);
PmResult pmBindBufferMemory(PmDevice device,
    PmBuffer buffer, PmDeviceMemory memory,
    PmDeviceSize offset);
```

Element formats [6.3]

```
typedef enum PmFormat {
    PM_FORMAT_UNDEFINED = 0,
    PM_FORMAT_R32_UINT = 1,
    PM_FORMAT_R32_SFLOAT = 2,
    PM_FORMAT_R32G32B32_SFLOAT = 3,
    PM_FORMAT_R32G32B32A32_SFLOAT = 4,
    PM_FORMAT_PARTICLE_STATE = 5,
    PM_FORMAT_RIGID_STATE = 6
} PmFormat;
typedef struct PmBufferView {
    PmBufferRange range;
    PmFormat format;
    uint32_t count;
    uint32_t stride;
} PmBufferView;
```

Triangle geometry [6.4]

```
typedef struct PmGeometryCreateInfo {
    PmStructureType sType;
    const void* pNext;
    PmBufferView vertices;
    PmBufferView indices;
    uint32_t queueFamilyIndex;
} PmGeometryCreateInfo;
PmResult pmCreateGeometry(PmDevice device,
    const PmGeometryCreateInfo* pInfo,
    PmGeometry* pGeometry);
void pmDestroyGeometry(PmDevice device,
    PmGeometry geometry);
void pmGetGeometryMemoryRequirements(
    PmDevice device, PmGeometry geometry,
    PmMemoryRequirements* pRequirements);
PmResult pmBindGeometryMemory(
    PmDevice device, PmGeometry geometry,
    PmDeviceMemory memory,
    PmDeviceSize offset);
```

Recorded geometry build [6.5]

```
PmResult pmCmdBuildGeometry(
    PmCommandBuffer commandBuffer,
    PmGeometry geometry);
```

Simulation state & rigid bodies [7]

Capacity-based simulation [7.1]

```
typedef struct PmFluidCapacity {
    uint32_t particleCapacity;
    uint32_t neighborCapacityPerParticle;
} PmFluidCapacity;
typedef struct PmSimulationCreateInfo {
    PmStructureType sType;
    const void* pNext;
    uint32_t queueFamilyIndex;
    uint32_t rigidBodyCapacity;
    uint32_t constraintCapacity;
    uint32_t contactCapacity;
    uint32_t fluidCount;
    const PmFluidCapacity* pFluidCapacities;
} PmSimulationCreateInfo;
```

Capacity-based simulation (continued) [7.1]

```
PmResult pmCreateSimulation(PmDevice device,
    const PmSimulationCreateInfo* pInfo,
    PmSimulation* pSimulation);
void pmDestroySimulation(PmDevice device,
    PmSimulation simulation);
```

Persistent & scratch bindings [7.2]

```
typedef struct
  PmSimulationMemoryRequirements {
    PmMemoryRequirements persistent;
    PmMemoryRequirements scratch;
} PmSimulationMemoryRequirements;
typedef struct PmSimulationMemoryBinding {
    PmDeviceMemory persistentMemory;
    PmDeviceSize persistentOffset;
    PmDeviceMemory scratchMemory;
    PmDeviceSize scratchOffset;
} PmSimulationMemoryBinding;
void pmGetSimulationMemoryRequirements(
    PmDevice device, PmSimulation
      simulation,
    PmSimulationMemoryRequirements* pInfo);
PmResult pmBindSimulationMemory(
    PmDevice device, PmSimulation
      simulation,
    const PmSimulationMemoryBinding*
      pBinding);
```

Portable rigid state record [7.3]

```
typedef struct PmRigidState {
    float position[3];
    float reserved0;
    float orientation[4];
    float linearVelocity[3];
    float reserved1;
    float angularVelocity[3];
    float reserved2;
} PmRigidState;
typedef enum PmMotionType {
    PM_MOTION_STATIC = 0,
    PM_MOTION_KINEMATIC = 1,
    PM_MOTION_DYNAMIC = 2
} PmMotionType;
```

Body descriptions [7.4]

```
typedef struct PmRigidBodyInfo {
    PmBool32 active;
    PmMotionType motionType;
    PmGeometry geometry;
    PmRigidState state;
    float mass;
    float inertiaDiagonal[3];
    float friction;
    float restitution;
    float linearDamping;
    float angularDamping;
    float collisionMargin;
} PmRigidBodyInfo;
PmResult pmCmdWriteRigidBodies(
    PmCommandBuffer commandBuffer,
    PmSimulation simulation,
    uint32_t firstBody, uint32_t count,
    const PmRigidBodyInfo* pBodies);
```

Forces, impulses & targets [7.5]

```
typedef struct PmRigidInput {
    uint32_t bodyIndex;
    float force[3];
    float torque[3];
    float linearImpulse[3];
    float angularImpulse[3];
} PmRigidInput;
PmResult pmCmdApplyRigidInputs(
    PmCommandBuffer commandBuffer,
    PmSimulation simulation,
    uint32_t count, const PmRigidInput*
      pInputs);
PmResult pmCmdSetRigidStates(
    PmCommandBuffer commandBuffer,
    PmSimulation simulation,
    uint32_t firstBody, uint32_t count,
    const PmRigidState* pStates);
```

Rigid constraints [8]

Joint families [8.1]

```
typedef enum PmConstraintType {
    PM_CONSTRAINT_FIXED = 0,
    PM_CONSTRAINT_POINT = 1,
    PM_CONSTRAINT_HINGE = 2,
    PM_CONSTRAINT_SLIDER = 3,
    PM_CONSTRAINT_PISTON = 4,
    PM_CONSTRAINT_GENERIC = 5,
    PM_CONSTRAINT_GENERIC_SPRING = 6,
    PM_CONSTRAINT_MOTOR = 7
} PmConstraintType;
typedef struct PmConstraintFrame {
    float position[3];
    float orientation[4];
} PmConstraintFrame;
```

Per-axis controls [8.2]

```
typedef struct PmAxisLimit {
    PmBool32 enabled;
    float lower;
    float upper;
} PmAxisLimit;
typedef struct PmAxisSpring {
    PmBool32 enabled;
    float stiffness;
    float damping;
    float equilibrium;
} PmAxisSpring;
typedef struct PmAxisMotor {
    PmBool32 enabled;
    float targetVelocity;
    float maximumForce;
} PmAxisMotor;
```

Constraint description [8.3]

```
typedef struct PmRigidConstraintInfo {
    PmBool32 active;
    PmConstraintType type;
    uint32_t bodyA;
    uint32_t bodyB;
    PmConstraintFrame frameA;
    PmConstraintFrame frameB;
    PmAxisLimit linearLimits[3];
    PmAxisLimit angularLimits[3];
    PmAxisSpring linearSprings[3];
    PmAxisSpring angularSprings[3];
    PmAxisMotor linearMotors[3];
    PmAxisMotor angularMotors[3];
    PmBool32 collideConnected;
} PmRigidConstraintInfo;
PmResult pmCmdWriteRigidConstraints(
    PmCommandBuffer commandBuffer,
    PmSimulation simulation,
    uint32_t firstConstraint, uint32_t
      count,
    const PmRigidConstraintInfo*
      pConstraints);
```

Degrees of freedom [8.4]

FIXED	All translation and rotation locked
POINT	Three translations locked; rotation free
HINGE	Only local X rotation eligible
SLIDER	Only local X translation eligible
PISTON	Local X translation and rotation eligible
GENERIC	All six axes eligible for limits/motors
GENERIC_SPRING	All six axes; limits, springs and motors
MOTOR	All axes free except driven/limited axes

Physics pipelines & materials [9]

Rigid solver state [9.1]

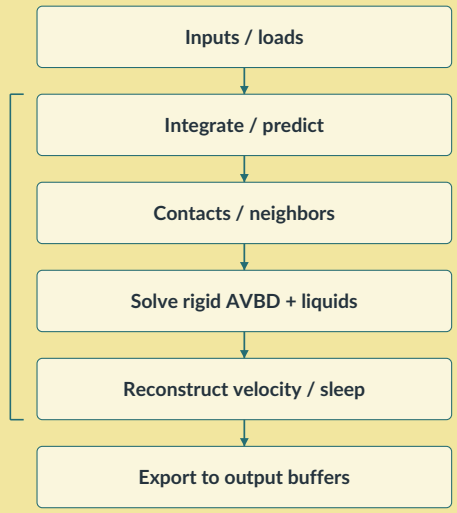
```
typedef struct PmRigidSolverState {
    uint32_t rigidIterations;
    PmBool32 enableSleeping;
    float sleepLinearThreshold;
    float sleepAngularThreshold;
    float sleepTimeSeconds;
} PmRigidSolverState;
```

Particle-liquid material [9.2]

```
typedef struct PmLiquidMaterialState {
    float restDensity;
    float particleRadius;
    float supportRadius;
    float restParticleVolume;
    float pressureStiffness;
    float kinematicViscosity;
    float velocityDamping;
    float maximumSpeed;
    uint32_t densityIterations;
} PmLiquidMaterialState;
```

Coupling state [9.3]

```
typedef enum PmCouplingType {
    PM_COUPLING_LIQUID_RIGID = 0,
    PM_COUPLING_LIQUID_LIQUID = 1
} PmCouplingType;
typedef struct PmCouplingState {
    PmCouplingType type;
    uint32_t fluidA;
    uint32_t fluidB;
    float contactDistance;
    float normalStiffness;
    float normalDamping;
    float tangentialDrag;
    float maximumForce;
    PmBool32 twoWay;
} PmCouplingState;
```



Substep loop

Pipeline creation [9.4]

```
typedef struct PmPhysicsPipelineCreateInfo {
    PmStructureType sType;
    const void* pNext;
    const PmRigidSolverState* pRigidState;
    uint32_t fluidMaterialCount;
    const PmLiquidMaterialState*
      pFluidMaterials;
    uint32_t couplingCount;
    const PmCouplingState* pCouplings;
} PmPhysicsPipelineCreateInfo;
PmResult pmCreatePhysicsPipeline(
    PmDevice device,
    const PmPhysicsPipelineCreateInfo*
      pInfo,
    PmPhysicsPipeline* pPipeline);
void pmDestroyPhysicsPipeline(
    PmDevice device,
    PmPhysicsPipeline pipeline);
```

Command recording & lifecycle [10]

Command pools [10.1]

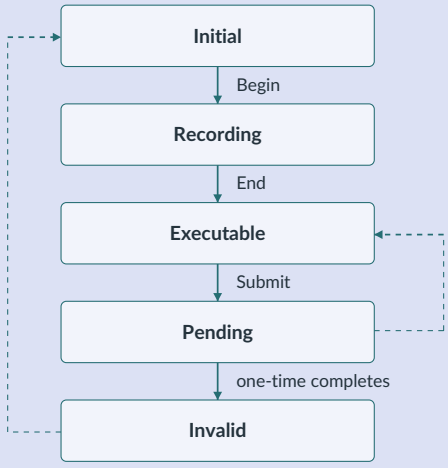
```
enum {
    PM_COMMAND_POOL_RESET_BUFFER_BIT = 0x1,
    PM_COMMAND_BUFFER_ONE_TIME_BIT = 0x1
};
typedef struct PmCommandPoolCreateInfo {
    PmStructureType sType;
    const void* pNext;
    PmFlags flags;
    uint32_t queueFamilyIndex;
} PmCommandPoolCreateInfo;
PmResult pmCreateCommandPool(PmDevice
  device,
    const PmCommandPoolCreateInfo* pInfo,
    PmCommandPool* pPool);
void pmDestroyCommandPool(PmDevice device,
    PmCommandPool pool);
PmResult pmResetCommandPool(PmDevice device,
    PmCommandPool pool);
```

Allocate, record & reset [10.2]

```
typedef struct PmCommandBufferBeginInfo {
    PmStructureType sType;
    const void* pNext;
    PmFlags flags;
} PmCommandBufferBeginInfo;
```

Allocate, record & reset (continued) [10.2]

```
PmResult pmAllocateCommandBuffers(
    PmDevice device, PmCommandPool pool,
    uint32_t count, PmCommandBuffer*
      pBuffers);
void pmFreeCommandBuffers(PmDevice device,
    PmCommandPool pool, uint32_t count,
    const PmCommandBuffer* pBuffers);
PmResult pmBeginCommandBuffer(
    PmCommandBuffer commandBuffer,
    const PmCommandBufferBeginInfo* pInfo);
PmResult pmEndCommandBuffer(
    PmCommandBuffer commandBuffer);
PmResult pmResetCommandBuffer(
    PmCommandBuffer commandBuffer);
```



Reusable completion → Executable

Reset → Initial; never while Pending

Legal states & resource lifetime [10.3]

Initial	Begin → Recording; may free/reset
Recording	pmCmd* only; End → Executable
Executable	QueueSubmit → Pending; may reset/free
Pending	No edits, reset, free or resubmit
Completion	Reusable → Executable; one-time → Invalid
Invalid	Reset → Initial, or free

Reset & particle initialization [10.4]

```
typedef struct PmParticleState {
    float position[3];
    float temperatureKelvin;
    float velocity[3];
    float reserved;
} PmParticleState;
PmResult pmCmdResetSimulation(
    PmCommandBuffer commandBuffer,
    PmSimulation simulation);
PmResult pmCmdSetFluidParticles(
    PmCommandBuffer commandBuffer,
    PmSimulation simulation,
    uint32_t fluidIndex,
    const PmBufferView* pParticles);
```

Bind & step [10.5]

```
typedef struct PmStepInfo {
    float timestep;
    uint32_t substeps;
    float gravity[3];
} PmStepInfo;
PmResult pmCmdBindPhysicsPipeline(
    PmCommandBuffer commandBuffer,
    PmPhysicsPipeline pipeline);
PmResult pmCmdStepSimulation(
    PmCommandBuffer commandBuffer,
    PmSimulation simulation,
    const PmStepInfo* pStep);
```

Data movement & output [11]

Buffer transfers [11.1]

```
typedef struct PmBufferCopy {
    PmDeviceSize srcOffset;
    PmDeviceSize dstOffset;
    PmDeviceSize size;
} PmBufferCopy;
```

Buffer transfers (continued) [11.1]

```
PmResult pmCmdCopyBuffer(
    PmCommandBuffer commandBuffer,
    PmBuffer src, PmBuffer dst,
    uint32_t count,
    const PmBufferCopy* pRegions);
```

Copy simulation output [11.2]

```
PmResult pmCmdCopyRigidStates(
    PmCommandBuffer commandBuffer,
    PmSimulation simulation,
    uint32_t firstBody, uint32_t count,
    const PmBufferRange* pDestination);
PmResult pmCmdCopyFluidParticles(
    PmCommandBuffer commandBuffer,
    PmSimulation simulation,
    uint32_t fluidIndex,
    const PmBufferRange* pDestination);
```

Submission & synchronization [12]

Stage and access scopes [12.1]

```
#define PM_STAGE_HOST_BIT UINT64_C(0x1)
#define PM_STAGE_TRANSFER_BIT UINT64_C(0x2)
#define PM_STAGE_UPDATE_BIT UINT64_C(0x4)
#define PM_STAGE_GEOMETRY_BUILD_BIT
  UINT64_C(0x8)
#define PM_STAGE_PHYSICS_BIT UINT64_C(0x10)
#define PM_STAGE_ALL_COMMANDS_BIT
  UINT64_C(0x1e)
#define PM_ACCESS_HOST_READ_BIT
  UINT64_C(0x1)
#define PM_ACCESS_HOST_WRITE_BIT
  UINT64_C(0x2)
#define PM_ACCESS_READ_BIT UINT64_C(0x4)
#define PM_ACCESS_WRITE_BIT UINT64_C(0x8)
```

Memory & buffer barriers [12.2]

```
typedef struct PmMemoryBarrier {
    PmStageFlags srcStages;
    PmAccessFlags srcAccess;
    PmStageFlags dstStages;
    PmAccessFlags dstAccess;
} PmMemoryBarrier;
typedef struct PmBufferMemoryBarrier {
    PmMemoryBarrier memory;
    PmBufferRange range;
    uint32_t srcQueueFamilyIndex;
    uint32_t dstQueueFamilyIndex;
} PmBufferMemoryBarrier;
PmResult pmCmdPipelineBarrier(
    PmCommandBuffer commandBuffer,
    uint32_t memoryBarrierCount,
    const PmMemoryBarrier* pMemoryBarriers,
    uint32_t bufferBarrierCount,
    const PmBufferMemoryBarrier*
      pBufferBarriers);
```

Timeline semaphores [12.3]

```
typedef struct PmSemaphoreCreateInfo {
    PmStructureType sType;
    const void* pNext;
    uint64_t initialValue;
} PmSemaphoreCreateInfo;
typedef struct PmSemaphoreSubmitInfo {
    PmSemaphore semaphore;
    uint64_t value;
    PmStageFlags stages;
} PmSemaphoreSubmitInfo;
PmResult pmCreateSemaphore(PmDevice device,
    const PmSemaphoreCreateInfo* pInfo,
    PmSemaphore* pSemaphore);
void pmDestroySemaphore(PmDevice device,
    PmSemaphore semaphore);
PmResult pmGetSemaphoreCounterValue(
    PmDevice device, PmSemaphore semaphore,
    uint64_t* pValue);
PmResult pmWaitSemaphore(PmDevice device,
    PmSemaphore semaphore, uint64_t value,
    uint64_t timeoutNanoseconds);
```

Queue submission [12.4]

```
typedef struct PmSubmitInfo {
    PmStructureType sType;
    const void* pNext;
    uint32_t waitCount;
    const PmSemaphoreSubmitInfo* pWaits;
    uint32_t commandBufferCount;
    const PmCommandBuffer* pCommandBuffers;
    uint32_t signalCount;
    const PmSemaphoreSubmitInfo* pSignals;
} PmSubmitInfo;
PmResult pmQueueSubmit(PmQueue queue,
    const PmSubmitInfo* pInfo, PmFence
      fence);
```

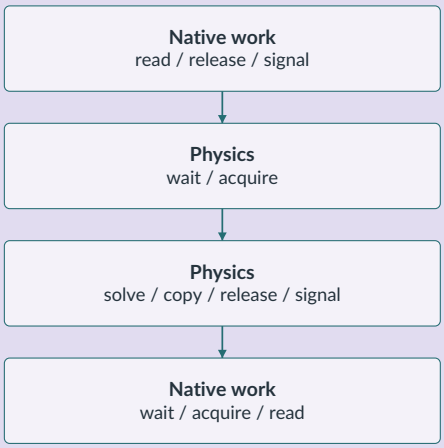
Queue submission (continued) [12.4]

```
PmResult pmQueueWaitIdle(PmQueue queue);
PmResult pmDeviceWaitIdle(PmDevice device);
```

Host completion fences [12.5]

```
enum { PM_FENCE_SIGNALED_BIT = 0x1 };
PmResult pmCreateFence(PmDevice device,
    PmFlags flags, PmFence* pFence);
void pmDestroyFence(PmDevice device,
    PmFence fence);
PmResult pmGetFenceStatus(PmDevice device,
    PmFence fence);
PmResult pmResetFences(PmDevice device,
    uint32_t count, const PmFence* pFences);
PmResult pmWaitForFences(PmDevice device,
    uint32_t count, const PmFence* pFences,
    PmBool32 waitAll, uint64_t
      timeoutNanoseconds);
```

Native API interoperability [13]



Timeline + resource handoff

Backend capabilities [13.1]

```
typedef enum PmNativeBackend {
    PM_NATIVE_BACKEND_CUDA = 1,
    PM_NATIVE_BACKEND_METAL = 2,
    PM_NATIVE_BACKEND_D3D12 = 3,
    PM_NATIVE_BACKEND_VULKAN = 4
} PmNativeBackend;
typedef uint64_t PmNativeHandle;
typedef struct PmNativeInteropProperties {
    PmBool32 deviceImport;
    PmBool32 bufferImport;
    PmBool32 timelineImport;
} PmNativeInteropProperties;
PmResult
  pmGetPhysicalDeviceInteropProperties(
    PmPhysicalDevice physicalDevice,
    PmNativeBackend backend,
    PmNativeInteropProperties* pProperties);
```

Device and queue import [13.2]

```
enum {
    PM_NATIVE_TIMELINE_ENABLED_BIT = 0x1,
    PM_NATIVE_SYNCHRONIZATION2_BIT = 0x2
};
typedef struct PmNativeQueueInfo {
    PmNativeHandle queue;
    uint32_t familyIndex;
    uint32_t queueIndex;
} PmNativeQueueInfo;
typedef struct PmNativeDeviceInfo {
    PmNativeBackend backend;
    PmNativeHandle instance;
    PmNativeHandle physicalDevice;
    PmNativeHandle device;
} PmNativeDeviceInfo;
typedef struct PmNativeDeviceImportInfo {
    PmStructureType sType;
    const void* pNext;
    PmNativeDeviceInfo nativeDevice;
    uint32_t queueCount;
    const PmNativeQueueInfo* pQueues;
    PmFlags enabledNativeFeatures;
    PmPhysicalDeviceFeatures
      enabledFeatures;
} PmNativeDeviceImportInfo;
PmResult pmImportNativeDevice(
    PmInstance instance,
    const PmNativeDeviceImportInfo* pInfo,
    PmDevice* pDevice);
```

Device and queue import (continued) [13.2]

```
PmResult pmGetNativeDevice(PmDevice device,
    PmNativeDeviceInfo* pNativeDevice);
```

Buffer import [13.3]

```
typedef struct PmNativeBufferInfo {
    PmNativeHandle buffer;
    PmDeviceSize offset;
    PmDeviceSize size;
    PmFlags usage;
    uint64_t nativeUsage;
    uint32_t owningQueueFamilyIndex;
} PmNativeBufferInfo;
PmResult pmImportNativeBuffer(PmDevice
  device,
    const PmNativeBufferInfo* pInfo,
    PmBuffer* pBuffer);
PmResult pmGetNativeBuffer(PmDevice device,
    PmBuffer buffer,
    PmNativeBufferInfo* pNativeBuffer);
```

Timeline import [13.4]

```
PmResult pmImportNativeSemaphore(PmDevice
  device,
    PmNativeHandle nativeSemaphore,
    PmSemaphore* pSemaphore);
PmResult pmGetNativeSemaphore(PmDevice
  device,
    PmSemaphore semaphore,
    PmNativeHandle* pNativeSemaphore);
```

Native object mapping [13.5]

CUDA	CUcontext / CUstream; CUdeviceptr; CUexternalSemaphore
Metal	MTLDevice / MTLCommandQueue; MTLBuffer; MTLSharedEvent
DirectX 12	ID3D12Device / CommandQueue; Resource; Fence
Vulkan	VkDevice / VkQueue; VkBuffer; timeline VkSemaphore

Typed Vulkan device adapter [13.6]

```
typedef struct PmVulkanQueueInfo {
    VkQueue queue;
    uint32_t familyIndex;
    uint32_t queueIndex;
} PmVulkanQueueInfo;
typedef struct PmVulkanDeviceInfo {
    VkInstance instance;
    VkPhysicalDevice physicalDevice;
    VkDevice device;
} PmVulkanDeviceInfo;
typedef struct PmVulkanDeviceImportInfo {
    VkInstance instance;
    VkPhysicalDevice physicalDevice;
    VkDevice device;
    uint32_t queueCount;
    const PmVulkanQueueInfo* pQueues;
    VkBool32 timelineSemaphoreEnabled;
    VkBool32 synchronization2Enabled;
    PmPhysicalDeviceFeatures
      enabledFeatures;
} PmVulkanDeviceImportInfo;
PmResult pmImportVulkanDeviceEXT(
    PmInstance instance,
    const PmVulkanDeviceImportInfo* pInfo,
    PmDevice* pDevice);
PmResult pmGetVulkanDeviceEXT(PmDevice
  device,
    PmVulkanDeviceInfo* pNativeDevice);
```

Typed Vulkan buffer adapter [13.7]

```
typedef struct PmVulkanBufferImportInfo {
    VkBuffer buffer;
    VkDeviceSize size;
    VkBufferUsageFlags nativeUsage;
    PmFlags usage;
    uint32_t owningQueueFamilyIndex;
} PmVulkanBufferImportInfo;
PmResult pmImportVulkanBufferEXT(
    PmDevice device,
    const PmVulkanBufferImportInfo* pInfo,
    PmBuffer* pBuffer);
PmResult pmGetVulkanBufferEXT(PmDevice
  device,
    PmBuffer buffer, VkBuffer*
      pNativeBuffer);
```

Typed Vulkan timeline adapter [13.8]

```
PmResult pmImportVulkanSemaphoreEXT(
    PmDevice device, VkSemaphore semaphore,
    PmSemaphore* pSemaphore);
PmResult pmGetVulkanSemaphoreEXT(
    PmDevice device, PmSemaphore semaphore,
    VkSemaphore* pNativeSemaphore);
```

Native synchronization [13.10]

CUDA	Stream order + external timeline waits/signals
Metal	Command buffers + shared events + resource hazards
DirectX 12	Queue fences + resource barriers
Vulkan	Timeline semaphores + memory/ownership barriers

Timing, work & memory queries [14]

Query pools [14.1]

```
typedef enum PmQueryType {
    PM_QUERY_TIMESTAMP = 0,
    PM_QUERY_STEP_STATISTICS = 1
} PmQueryType;
typedef struct PmQueryPoolCreateInfo {
    PmStructureType sType;
    const void* pNext;
    PmQueryType queryType;
    uint32_t queryCount;
} PmQueryPoolCreateInfo;
PmResult pmCreateQueryPool(PmDevice device,
    const PmQueryPoolCreateInfo* pInfo,
    PmQueryPool* pPool);
void pmDestroyQueryPool(PmDevice device,
    PmQueryPool pool);
PmResult pmCmdResetQueryPool(
    PmCommandBuffer commandBuffer,
    PmQueryPool pool, uint32_t first,
    uint32_t count);
```

GPU timestamps [14.2]

```
PmResult pmCmdWriteTimestamp(
    PmCommandBuffer commandBuffer,
    PmStageFlags stage,
    PmQueryPool pool, uint32_t query);
```

Step statistics & execution status [14.3]

```
typedef struct PmStepStatistics {
    uint64_t stepCount;
    uint64_t substepsExecuted;
    uint64_t poseIterations;
    uint64_t impactIterations;
    uint64_t peakContactCount;
    uint64_t peakActiveBodies;
    uint64_t peakAwakeBodies;
    uint64_t peakParticleCount;
    uint64_t scratchHighWaterBytes;
    uint64_t overflowCount;
    int32_t executionResult;
    uint32_t reserved;
} PmStepStatistics;
PmResult pmCmdBeginStatisticsQuery(
    PmCommandBuffer commandBuffer,
    PmQueryPool pool, uint32_t query);
PmResult pmCmdEndStatisticsQuery(
    PmCommandBuffer commandBuffer,
    PmQueryPool pool, uint32_t query);
PmResult pmGetSimulationStatus(
    PmDevice device, PmSimulation
      simulation,
    PmResult* pExecutionResult);
```

Readback & availability [14.4]

```
enum {
    PM_QUERY_RESULT_WAIT_BIT = 0x1,
    PM_QUERY_RESULT_WITH_AVAILABILITY_BIT =
      0x2
};
PmResult pmGetQueryPoolResults(PmDevice
  device,
    PmQueryPool pool, uint32_t first,
    uint32_t count, size_t dataSize,
    void* pData, PmDeviceSize stride,
    PmFlags flags);
```

Device budget & library accounting [14.5]

```
typedef struct PmMemoryBudget {
    uint32_t heapCount;
    PmDeviceSize budget[16];
    PmDeviceSize usage[16];
    PmDeviceSize applicationAllocationBytes;
    PmDeviceSize libraryDeviceBytes;
    PmDeviceSize libraryHostBytes;
```

Device budget & library accounting (continued) [14.5]

```
    PmDeviceSize borrowedBufferBytes;
} PmMemoryBudget;
PmResult pmGetDeviceMemoryBudget(
    PmDevice device, PmMemoryBudget*
      pBudget);
```

Features, limits & formats [15]

Physical-device limits [15.1]

```
typedef struct PmPhysicalDeviceLimits {
    PmDeviceSize maxAllocationSize;
    PmDeviceSize maxBufferSize;
    PmDeviceSize nonCoherentAtomSize;
    uint32_t maxMemoryAllocations;
    uint32_t maxRigidBodies;
    uint32_t maxRigidConstraints;
    uint32_t maxContacts;
    uint32_t maxFluidSystems;
    uint32_t maxParticlesPerFluid;
    uint32_t maxNeighborsPerParticle;
    uint32_t maxRigidIterations;
    uint32_t maxLiquidIterations;
    uint32_t maxSubsteps;
    uint32_t maxQueriesPerPool;
    uint64_t maxTimelineValueDifference;
} PmPhysicalDeviceLimits;
void pmGetPhysicalDeviceLimits(
    PmPhysicalDevice physicalDevice,
    PmPhysicalDeviceLimits* pLimits);
```

Format support query [15.2]

```
enum {
    PM_FORMAT_GEOMETRY_VERTEX_BIT = 0x1,
    PM_FORMAT_GEOMETRY_INDEX_BIT = 0x2,
    PM_FORMAT_PARTICLE_INPUT_BIT = 0x4,
    PM_FORMAT_SIMULATION_OUTPUT_BIT = 0x8
};
typedef struct PmFormatProperties {
    uint32_t elementSize;
    uint32_t elementAlignment;
    PmFlags supportedUsages;
} PmFormatProperties;
void pmGetPhysicalDeviceFormatProperties(
    PmPhysicalDevice physicalDevice,
    PmFormat format,
    PmFormatProperties* pProperties);
```

Portable buffer layouts [15.3]

R32_UINT	4 bytes / alignment 4 / unsigned index
R32_SFLOAT	4 bytes / alignment 4 / IEEE binary32
R32G32B32_SFLOAT	12 bytes / alignment 4 / x,y,z
R32G32B32A32_SFLOAT	16 bytes / alignment 4 / x,y,z,w
PARTICLE_STATE	32 bytes / alignment 4 / PmParticleState
RIGID_STATE	64 bytes / alignment 4 / PmRigidState

Dependency checklist [15.4]

Rigid contacts	rigidBodies
Rigid joints	rigidBodies + rigidConstraints
One liquid	particleLiquids
Liquid ↔ bodies	both domains + liquidRigidCoupling
Liquid ↔ liquid	particleLiquids + liquidLiquidCoupling
GPU timing	timestampQueries + queue timestamp bits
Solver counters	statisticsQueries
Native object sharing	Per-backend import capabilities
Typed Vulkan adapters	vulkanInterop + required native features

Fundamental declarations [17]

Types, handles & constants [17.1]

```
#define PM_API_VERSION_0_2 ((uint32_t)(2u <<
  12))
#define PM_WHOLE_SIZE UINT64_MAX
#define PM_TIMEOUT_INFINITE UINT64_MAX
#define PM_INVALID_INDEX UINT32_MAX
#define PM_QUEUE_FAMILY_IGNORED UINT32_MAX
#define PM_QUEUE_FAMILY_EXTERNAL (UINT32_MAX
  - 1u)
#define PM_FALSE 0u
#define PM_TRUE 1u
typedef uint32_t PmBool32;
typedef uint32_t PmFlags;
typedef uint64_t PmDeviceSize;
typedef uint64_t PmStageFlags;
typedef uint64_t PmAccessFlags;
typedef struct PmInstance_T* PmInstance;
typedef struct PmPhysicalDevice_T*
  PmPhysicalDevice;
typedef struct PmDevice_T* PmDevice;
typedef struct PmQueue_T* PmQueue;
typedef struct PmDeviceMemory_T*
  PmDeviceMemory;
typedef struct PmBuffer_T* PmBuffer;
typedef struct PmGeometry_T* PmGeometry;
typedef struct PmSimulation_T* PmSimulation;
typedef struct PmPhysicsPipeline_T*
  PmPhysicsPipeline;
typedef struct PmCommandPool_T*
  PmCommandPool;
typedef struct PmCommandBuffer_T*
  PmCommandBuffer;
typedef struct PmSemaphore_T* PmSemaphore;
typedef struct PmFence_T* PmFence;
typedef struct PmQueryPool_T* PmQueryPool;
```

Structure types [17.2]

```
typedef enum PmStructureType {
    PM_STRUCTURE_TYPE_UNDEFINED = 0,
    PM_STRUCTURE_TYPE_INSTANCE_CREATE_INFO =
      1,
    PM_STRUCTURE_TYPE_DEVICE_CREATE_INFO =
      2,
    PM_STRUCTURE_TYPE_MEMORY_ALLOCATE_INFO =
      3,
    PM_STRUCTURE_TYPE_BUFFER_CREATE_INFO =
      4,
    PM_STRUCTURE_TYPE_GEOMETRY_CREATE_INFO =
      5,
    PM_STRUCTURE_TYPE_SIMULATION_CREATE_INFO
    = 6,
PM_STRUCTURE_TYPE_PHYSICS_PIPELINE_CREATE_INFO
= 7,
PM_STRUCTURE_TYPE_COMMAND_POOL_CREATE_INFO =
  8,
PM_STRUCTURE_TYPE_COMMAND_BUFFER_BEGIN_INFO
 = 9,
    PM_STRUCTURE_TYPE_SEMAPHORE_CREATE_INFO
     = 10,
    PM_STRUCTURE_TYPE_SUBMIT_INFO = 11,
PM_STRUCTURE_TYPE_NATIVE_DEVICE_IMPORT_INFO
 = 12,
    PM_STRUCTURE_TYPE_QUERY_POOL_CREATE_INFO
    = 13,
} PmStructureType;
```

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PmAxisMotor 2
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